

MD. TANVIR AHMED TONMOY

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🌐 LinkedIn 📁 Portfolio

Research Focus

Compute-in-Memory; Neuromorphic Computing; Memory-Based AI Accelerators; Digital Hardware Design.

Education

BRAC University, Dhaka, Bangladesh
Bachelor of Science in Electrical and Electronic Engineering
Specialization: Electronics | CGPA: **3.72/4.00**

Publications

- H. H. Jui, **M. T. A. Tonmoy**, M. R. Kabbo, A. A. Mahi, N. M. Himel, S. Sultana, T. Titirsha, and T. Rahman, “Variation-Aware SRAM Compute-in-Memory Boolean Architecture for Binary Neural Networks,” accepted in *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, 2026, Kolkata, India.
- F. Uddin, N. Shahrair, M. M. Rahman, **M. T. A. Tonmoy**, S. Akter, M. Jobayer, M. K. Rahman, and M. M. H. Shawon, “Soft Robotic Bellow-Controlled Arm Using Multi-DOF Control System,” accepted in *5th Interdisciplinary Conference on Electrics and Computer (INTCEC) 2025*, Chicago, USA.
- M. M. Chowdhury, M. M. Rahman, M. Mahmud, K. A. Rahim, **M. T. A. Tonmoy**, A. Nawrin, C. D. Joy, R. S. I. Antara, and A. H. Kafi, “BRACU Diganta: An Affordable CanSat Learning Kit for Students in Developing Countries,” accepted in *International Astronautical Congress (IAC/IAF)*, 2025, Sydney, Australia.

On-Going Work

Synapse–Neuron Co-Regulated SRAM-CIM with Temporal-Criticality-Aware Verification for Robust SNN Inference Current

- Ongoing research on an 8T SRAM-based compute-in-memory architecture for SNNs, investigating PVT-aware synapse–neuron co-regulation and temporal reliability of LIF neuron dynamics.
- Exploring selective re-evaluation of timing-critical neuron states to improve spike-time robustness with minimal energy overhead.

Progressive Binary Hashing with Work-Conserving Dual-Port SRAM for Early-Exit Hamming Search Current

- Developing an algorithm–hardware co-design framework that learns binary representations optimized for progressive Hamming-distance evaluation and early candidate rejection.
- Designing a dual-port SRAM-based search accelerator with work-conserving scheduling, with FPGA validation targeting improved throughput, memory utilization, and energy efficiency.

Research Experience

SRAM Compute-in-Memory and Neuromorphic Hardware 2026–Present

- **Variation-Aware SRAM-CIM Boolean Architecture.** Designed and simulated a 45 nm 6T SRAM-based compute-in-memory Boolean architecture in Cadence Virtuoso. Implemented Boolean operations including AND, OR, NAND, NOR, XOR, and XNOR, and performed Monte Carlo delay analysis to evaluate timing-induced logic error probability. Developed a hardware-aware BNN evaluation framework in Python/PyTorch to connect circuit-level timing variability with inference accuracy.
- **8T SRAM Crossbar and SNN Architecture.** Currently designing and characterizing an 8T SRAM cell and developing a 4×4 crossbar architecture for array-level memory operation. Extending the architecture toward an SRAM-based spiking neural network for neuromorphic computing applications and evaluating circuit- and architecture-level performance.

Multi DOF Soft Robotics Rehabilitation Device 2024–2025

- **Pneumatic Bellows-Based Multi-DOF Rehabilitation Arm.** Worked on a soft robotic rehabilitation system using industrial bellows as pneumatic actuators for multi-DOF motion. Integrated sensing and control concepts including pressure sensing, IMU feedback, and trajectory tracking for rehabilitation-oriented assistive robotics.

FYDP/Capstone Project 2025–2026

- **Autonomous Agricultural Rover for Soil Sampling and Zone-Wise Fertilization.** Developed a precision agriculture rover concept involving grid-based soil sampling, NPK analysis, zone-wise fertilization, and simulation-based evaluation in Webots/Gazebo. Worked on system design, sensor integration, motion planning, and performance evaluation.

Course and Other Projects

45 nm CMOS Phase-Locked Loop Design in Cadence

2026

- Designed and simulated a CMOS PLL using Cadence Virtuoso and Maestro, including PFD, charge pump, loop filter, VCO, and divide-by-48 frequency divider. Performed DC and transient analysis and evaluated locked-state behavior, current consumption, and frequency response.

VHDL–FPGA Based RISC-V Processor Design

2026

- Designing and implementing a simplified RISC-V processor using synthesizable VHDL for FPGA implementation. Developing modular VHDL components and a comprehensive testbench to verify correct instruction execution through simulation and waveform analysis.

SRAM-Based In-Memory Computing Architecture in VHDL

2026

- Designed and simulated an SRAM-based in-memory computing architecture using VHDL in Xilinx ISE. Developed the SRAM memory structure, computational logic, control circuitry, and finite-state machine (FSM) required to coordinate memory and in-memory computing operations. Verified functionality using VHDL testbenches, simulation waveforms, and synthesized hardware schematics.

BRACU Diganta CanSat and Model Satellite Systems

2024–2026

- Worked on onboard electronics and system development for BRACU Diganta, selected among the global finalists of CanSat 2025 in Virginia, USA. Designed and integrated sensor interfaces, telemetry electronics, beacon logic, and payload electronics.
- Also leading the electronics subsystem for the Teknofest 2026 model satellite competition, including science payload electronics, flight controller prototyping, communication subsystem planning, and system integration.

CNN-Based Floating Waste Classification on Water Surfaces

2026

- Implemented a CNN-based visual recognition pipeline for classifying floating waste materials on water surfaces into six categories: algae, chemical, mixed, natural, plastic, and polythene pollution. Worked on dataset preprocessing, image augmentation, CNN training, validation analysis, and qualitative testing for autonomous water-cleaning system applications.

Awards and Achievements

- Merit Based Scholarship** – Awarded a 25% tuition scholarship for academic excellence for four semesters and 10% for three semesters.
- Dean's List Award** – Recognized on the Dean's List for outstanding academic performance for six semesters.
- CanSat Competition 2025 Global Finalist** – American Astronautical Society, Virginia, USA.
- Industrial Attachment, Neural Semiconductor Ltd.** – Gained exposure to semiconductor design workflows including digital design, physical design, analog design, RF design, and fabrication-related processes.
- Undergraduate Teaching Assistant** – Served as an undergraduate teaching assistant for one semester in the Department of Electrical and Electronic Engineering, BRAC University.
- International Conference Publications** – ISVLSI 2026, INTCEC 2025, and IAC/IAF 2025.

Leadership and Community Involvement

Electronics Subsystem Lead, BRACU Diganta CanSat Team

Led onboard electronics design, sensor integration, power system planning, and telemetry-related development.

Research and Technical Contributor

Contributed to interdisciplinary projects in VLSI, soft robotics, space systems, embedded systems, and precision agriculture.

References

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